CariSurg MedTech Pathways Research Project Week 4 Final Submission: Ethics, Safety & Risk Awareness in Healthcare Technology

Project Title: AI-Assisted Risk Flagging for Emergency Department Triage in Resource-Constrained Settings
Student Name: Vishal Baboolal

Acronym Guide

AI – Artificial Intelligence
ED – Emergency Department
EHR – Electronic Health Record
EMS – Emergency Medical Services
ESI – Emergency Severity Index
KTS – Kampala Trauma Score
LLM – Large Language Model **ML**
– Machine Learning

1. Statement of the Problem

Emergency departments often deal with overcrowding, time pressure, and variation in triage decisions, especially when nurses must quickly identify patients who may be at higher risk (Da'Costa et al., 2025; De Freitas et al., 2020; Tyler et al., 2024). Although recent studies show that artificial intelligence (AI) can support emergency triage, many tools are developed using retrospective or single-site datasets and do not always show how the output would fit into real bedside decision-making (Agius et al., 2025; Tahernejad et al., 2024; Wong et al., 2021).

The key problem is that it remains unclear how a simple, nurse-facing AI risk flag could be validated and integrated into triage workflows in a resource-constrained emergency department without increasing alert burden or replacing clinical judgement. For the tool to be clinically useful, it must fit into the existing emergency department workflow, respect nurse time pressure, handle missing or delayed data, and provide information at the point where triage decisions are actually made.

2. Previous Work and Literature Review

(Tyler et al., 2024) reviewed the use of AI in hospital emergency department triage. The problem addressed was the increasing pressure on emergency departments and the need to identify whether AI could support triage decisions. The method was a scoping review of studies that applied AI to emergency department triage. The review found that AI tools may help identify high-risk patients earlier, improve prioritisation, and support resource allocation. However, the authors also showed that many studies focus heavily on model performance, with less attention to workflow integration and real-world clinical use. This supports the need for a pilot that tests not only whether a model can perform well, but whether a nurse can actually use the output during triage (Tyler et al., 2024).

(Da'Costa et al., 2025)reviewed AI-driven triage in emergency departments. The problem addressed was whether AI could reduce delays, improve prioritisation, and support overloaded emergency care systems. The review discussed potential benefits such as faster patient sorting, reduced waiting times, and improved use of limited resources. However, it also identified challenges including data quality, bias, clinician trust, ethical concerns, and implementation barriers. This paper is relevant because it supports the idea that an AI triage tool must be designed as a decision-support system rather than a replacement for clinical judgement (Da'Costa et al., 2025)

(Araouchi & Adda, 2024) presented TriageIntelli, an AI-assisted multimodal triage system for health centres. The problem addressed was the need to support triage classification using multiple types of patient data. The study compared several machine learning models, including random forests, support vector machines, artificial neural networks, gradient boosting, logistic regression, XGBoost, and a stacking model. The outcome suggested that machine learning can predict triage categories with promising accuracy. However, the study also highlights a limitation that is important for this proposal: accuracy alone does not prove that a tool is safe, understandable, or usable in a busy emergency department workflow (Araouchi & Adda, 2024).

(Agius et al., 2025) used emergency department data from Mater Dei Hospital in Malta to predict Emergency Severity Index level, hospital admission, and admitting ward. The problem addressed was whether routinely collected emergency department data could support triage and resource planning. The method involved XGBoost-based machine learning models. The outcome showed that emergency department data can support prediction of triage-related outcomes, and that performance may improve when more clinical information is included. The limitation for this proposal is that findings from one hospital context cannot be assumed to transfer directly to a Caribbean or resourceconstrained emergency department without local validation (Agius et al., 2025).

(Tahernejad et al., 2024)reviewed the application of AI in triage during emergencies and disasters. The problem addressed was how AI could support triage, response time, resource allocation, and emergency decision-making in high-pressure settings. The review found that AI may help with real-time data transfer, vital sign monitoring, and resource management. However, the authors also identified barriers such as staff training, trust, equipment limitations, privacy concerns, and implementation challenges. This is useful for the Mercer ED scenario because the proposed tool must be simple, realistic, and designed around actual workflow limitations (Tahernejad et al., 2024).

(De Freitas et al., 2020) explored patient flow in a Caribbean emergency department. The problem addressed was how patient flow, overcrowding, and process bottlenecks affect emergency care. The method was a qualitative exploration of patient flow in a Caribbean emergency department. The study highlighted delays, resource limitations, and workflow pressures across the patient journey. Although this paper is not a recent AI-triage paper, it is valuable because it provides regional and workflow context. It supports the argument that an AI triage tool for Mercer General must be designed for real emergency department constraints rather than only for technical performance (De Freitas et al., 2020)

(Wong et al., 2021) externally validated a widely implemented sepsis prediction model in hospitalized patients. The problem addressed was whether a widely used proprietary prediction model performed well outside the setting in which it was developed. The study found that real-world performance was weaker than expected. The limitation highlighted is important: prediction tools can perform differently when deployed in new clinical settings, and poor calibration or alert burden can affect clinical usefulness. This supports the need for local validation, careful monitoring, and human oversight before using AI-assisted risk alerts in emergency triage (Wong et al., 2021)

(Mitchell et al., 2024)described triage implementation in resource-limited emergency departments in the Pacific region. The problem addressed was how to implement triage systems in settings where staff, infrastructure, training, and data systems may be limited. The article described four implementation strategies: needs assessment, digital learning, public communications, and electronic data management. The outcome was a practical set of implementation lessons from the Pacific region, including the value of understanding local needs and using simple electronic registries to support triage quality improvement. This paper is important for the proposed Mercer pilot because it shows that triage improvement is not only a technical issue; it also depends on workflow redesign, staff training, public understanding, and data systems (Mitchell et al., 2024).

(Masanneck et al., 2024)compared the triage performance of large language models, ChatGPT, trained emergency department staff, and untrained doctors. The problem addressed was whether large language models could perform emergency triage safely and effectively. The study used emergency department case vignettes and compared triage performance across different raters. The outcome showed that large language models and ChatGPT did not yet match professionally trained raters, and the models tended toward overtriage. This is useful for the proposed pilot because it supports the need to keep the triage nurse as the final decision-maker and to treat AI output as a support signal rather than an automated triage decision (Masanneck et al., 2024).

(Nsubuga et al., 2025) studied machine learning for trauma triage in a low-resource setting and compared machine learning models with the Kampala Trauma Score. The problem addressed was whether machine learning could improve trauma triage prediction where traditional triage tools may have limited predictive accuracy. The study used data from 4,109 trauma patients at Soroti Regional Referral Hospital in rural Uganda and trained four machine learning models. The outcome showed that all four machine learning models outperformed the Kampala Trauma Score, with Random Forest and Gradient Boosting achieving strong AUC-ROC values. However, the authors stated that further validation is needed before integration into clinical practice. This paper supports the proposal because it shows that machine learning may help triage in low-resource settings, but validation and careful implementation remain necessary (Nsubuga et al., 2025).

(Ciecierski-Holmes et al., 2022)conducted a systematic scoping review of artificial intelligence applications for strengthening healthcare systems in low- and middle-income countries. The problem addressed was whether AI technologies have been evaluated in real-world LMIC healthcare settings. The review found that only a small number of evaluated AI implementations existed, and that applications included clinical decision support, treatment planning, triage assistants, and health chatbots. However, the review also identified major challenges, including reliability issues, mixed workflow effects, poor user-friendliness, limited transparency about training data and algorithms, and lack of adaptation to local context. This paper is important for the Mercer proposal because it supports the need for local validation, workflow testing, and infrastructure-aware implementation before any AI triage tool is scaled across a public hospital network (Ciecierski-Holmes et al., 2022)

(Mallon et al., 2025) studied artificial intelligence in prehospital emergency care systems in low- and middle-income countries through a scoping review and expert consultation. The problem addressed was whether AI could support emergency care before hospital arrival in lower-resource settings. The study identified themes around AI development, resource requirements, data considerations, human factors, and local context. The conclusion was that AI may support emergency care, but only if infrastructure, staff education, data quality, ethical concerns, and local context are addressed. This is relevant to the proposed AI-assisted triage tool because one of the proposed plug-in points uses EMS handover information before full registration. The paper supports the need to treat pre-triage AI as a cautious early-warning support, not as an autonomous decision-maker (Mallon et al., 2025)

(Aljubran et al., 2025) examined the use of machine learning algorithms to enhance paediatric emergency department triage. The problem addressed was that conventional paediatric triage systems may rely heavily on subjective assessment, while machine learning may help classify paediatric patients into urgency groups. The study developed a model to classify patients into non-urgent, urgent, and emergency categories, with CatBoost reported as the strongest performing model. However, the authors noted important limitations, including that the dataset came from one centre, historical data may not reflect changing patient profiles or triage practices, and the dataset did not include clinical outcomes from index visits. This paper is relevant because it supports the equity concern that paediatric patients should not be assumed to be safely covered by a general adult-focused triage model without subgroup validation (Aljubran et al., 2025).

(Fuller et al., 2023) trained and tested a gradient boosted machine learning model to predict adverse outcomes in patients presenting to emergency departments with suspected COVID-19 infection in a middle-income setting. The study used linked routine data from public-sector emergency departments in the Western Cape, South Africa, and externally validated the model in later Western Cape, United Kingdom, and Sudanese cohorts. The model performed strongly in internal testing, but its performance was reduced during temporal and geographical external validation. This paper is important for the Mercer proposal because it shows that even a well-developed emergency department prediction model can perform differently across time, countries, disease waves, and health systems. It directly supports the risk register entry on distribution shift and the need for silent validation before site-wide rollout (Fuller et al., 2023).

(Ethics and Governance of Artificial Intelligence for Health, 2021)published guidance on the ethics and governance of artificial intelligence for health. The guidance argues that ethics and human rights must be placed at the centre of AI design, deployment, and use. It identifies key governance concerns including transparency, accountability, equity, patient and community impact, and responsible oversight. This report is relevant because the Mercer proposal is not only a technical model-building exercise; it is a proposed clinical decision-support system that would affect patients, nurses, and hospital workflows. The WHO guidance supports the proposal's human-in-the-loop stance, patient-facing disclosure, named clinical accountability, and the requirement that the tool should support rather than replace clinical judgement (Ethics and Governance of Artificial Intelligence for Health, 2021)

3. Gaps Identified Gap 1: Limited Local Validation in Resource-Constrained Emergency Departments

Many AI triage studies report promising results, but the models are often developed using retrospective or single-hospital data. This makes it difficult to know whether the same approach would work well in another hospital, especially in a resource-constrained Caribbean emergency department where patient flow, staffing, data quality, and system pressures may differ from larger international hospitals.

This gap is important because emergency department AI tools should not be accepted based only on model performance in another setting. A 12-week Mercer pilot can begin to address this by testing a simple risk-flagging approach using approved de-identified, simulated, or Mercer-style triage data before considering wider use.

Gap 2: Limited Evidence on Workflow Integration and Clinical Usefulness

A second gap is the difference between model performance and real clinical usefulness. Several studies report accuracy, AUC, or model comparisons, but they do not always explain how a nurse would use the output during triage. In practice, the tool must be clear, quick to interpret, and safe to use during interruptions, overcrowding, and time pressure.

This pilot can address that by designing the output as a simple urgent-review risk flag, along with the abnormal findings that triggered the flag, rather than trying to replace the triage nurse or automatically assign a final triage category.

4. Proposed Solution

The proposed solution is a 12-week pilot of a simple AI-assisted risk-flagging tool for emergency department triage. The tool would use information that is normally available during triage, such as vital signs, age, presenting complaint, and basic clinical observations. Its purpose would be to highlight patients who may need urgent review, not to replace the triage nurse or make final clinical decisions.

Based on the workflow map, the strongest location for the tool is after vital-sign capture and during the triage nurse assessment. At this point, the system would have enough information to check for abnormal values and generate a short risk flag, while still being early enough to support the nurse's triage decision. A secondary use would be an end-oftriage deterioration screen, where the system reviews the completed triage data and flags patients who may need earlier review.

The proposed output would be simple and nurse-facing. Instead of assigning a final triage category, the tool would show:

1. A risk flag, such as "urgent review suggested"
2. The main abnormal findings that triggered the flag
3. A missing-data warning if required inputs are incomplete
4. A timestamp showing when the output was generated

The pilot would use de-identified historical triage data if approved and available. If approved local data are not available within the 12-week timeframe, the prototype could first be tested using simulated or Mercer-style triage data based on realistic emergency department variables. This would allow the project to test the workflow, safety rules, missing-data handling, and risk-flagging logic without making live clinical decisions.

The pilot would begin by deciding which input variables and safety rules should be included. The data would then be cleaned and checked for missing, unrealistic, or inconsistent values. A simple baseline model could be developed and compared with basic rule-based clinical thresholds. The output would be kept easy to understand by showing the patient's risk flag and the main abnormal findings that contributed to it.

5. Existing Mercer ED Triage Workflow

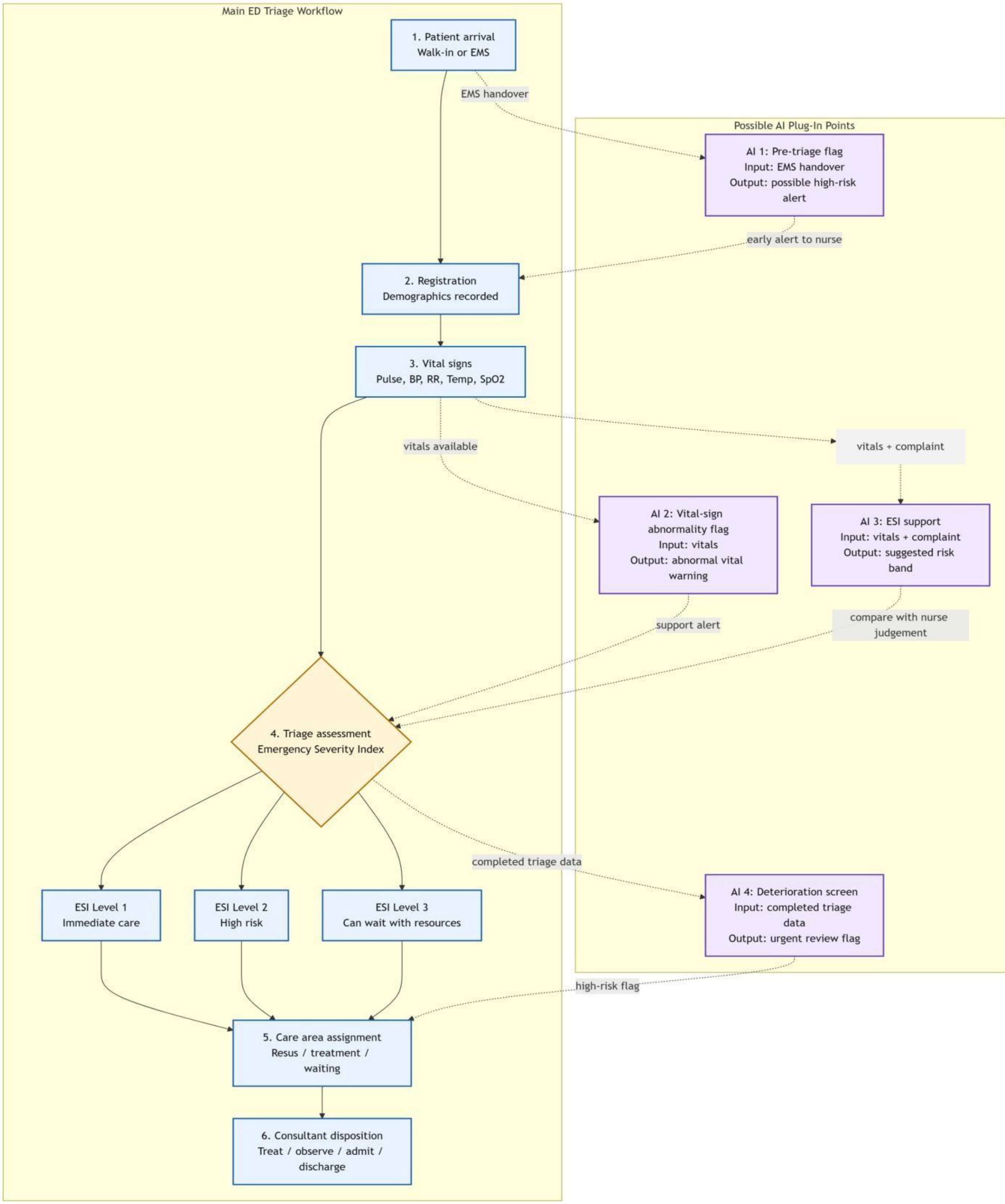
The current Mercer General emergency department triage process begins when the patient arrives either as a walk-in patient or by emergency medical services. If the patient arrives by ambulance, information may enter the system before registration through an ambulance radio report or handover. This early information may include the presenting complaint, urgency, vital concerns, or signs of deterioration.

After arrival, the patient moves to registration, where demographic and administrative details are recorded. The patient then proceeds to vital signs capture, where the nurse records measurements such as pulse, blood pressure, respiratory rate, temperature, oxygen saturation, and basic observations. In a busy or resource-constrained emergency department, this step may sometimes be paper-first, meaning the information is written down before being entered into the electronic health record.

The triage nurse then completes the triage assessment and assigns a triage level, such as Emergency Severity Index Level 1, Level 2, or Level 3. This decision affects how quickly the patient is seen and where the patient is sent next. The patient is then assigned to the appropriate care area, such as urgent review, resuscitation, treatment, observation, or waiting area. The care team or consultant later makes the disposition decision, which may include treatment, observation, admission, discharge, or transfer.

Important handoffs occur between emergency medical services and the emergency department team, registration and nursing staff, the triage nurse and care area staff, and finally the care team and consultant. Information can be delayed or lost when vital signs are documented on paper before electronic entry, when the department is crowded, or when staff are interrupted during the triage process.

6. Workflow Diagram Figure 1. Current Mercer ED triage workflow with AI plug-in points



The diagram should show the main triage pathway from patient arrival to consultant disposition. The main patient flow should be visually distinct from the AI plug-in points. The diagram should include the following AI integration points:

1. Pre-triage flag using emergency medical services handover information
2. Vital-sign abnormality flag after vital signs are captured
3. Emergency Severity Index support during triage nurse assessment
4. End-of-triage deterioration screen after the full assessment is completed

7. AI Plug-In Points Plug-In Point 1: Pre-Triage Flag

This plug-in point would sit at the patient arrival stage, especially when emergency medical services or ambulance radio information is available before the patient is fully registered. The AI system could review the early handover information and flag possible high-risk cases before the patient reaches full triage.

Input: Emergency medical services radio report or short ambulance handover
Output: Possible Level 1 or high-risk alert
Human action: The nurse prepares early review or alerts the care team if the patient appears unstable

Plug-In Point 2: Vital-Sign Abnormality Flag

This plug-in point would sit immediately after vital signs are recorded. The AI system could check pulse, oxygen saturation, blood pressure, respiratory rate, and temperature for abnormal or concerning values.

Input: Measured vital signs
Output: Abnormal vital-sign warning
Human action: The triage nurse checks the patient immediately and decides whether faster assessment is needed

Plug-In Point 3: Emergency Severity Index Support

This plug-in point would sit during the triage nurse's Emergency Severity Index assessment. The AI system could combine vital signs and presenting complaint information to suggest a risk band or highlight possible under-prioritisation.

Input: Vital signs and presenting complaint
Output: Suggested risk band or triage support alert

Human action: The triage nurse compares the AI suggestion with their own clinical judgement and makes the final decision

Plug-In Point 4: End-of-Triage Deterioration Screen

This plug-in point would sit after the full triage assessment. The AI system could review the completed triage data and flag patients who may need urgent review even if they were not initially identified as the highest priority.

Input: Full triage data
Output: Urgent review risk flag
Human action: The nurse or care team routes the patient earlier if the risk flag is high

8. Constraints & Stakeholders

8.1 Workflow Constraints

Constraint	Mechanism	Design Impact	Possible Mitigation
Electronic health record data lag	Vital signs may be written on paper or collected before being entered into the electronic health record. This creates a delay between the nurse measuring the patient and the AI system receiving usable data.	If the AI tool depends only on electronic health record data, the alert may appear too late to support the actual triage decision.	Allow quick manual entry or confirmation of key vital signs at the triage station. Show a timestamp for the AI output and warn the nurse if the data are incomplete or outdated.
Limited triage nurse time	Triage nurses work under time pressure, interruptions, overcrowding, and competing documentation demands.	If the AI tool adds extra clicks, long explanations, or complicated screens, it may increase cognitive load and be ignored during busy periods.	Keep the output short and actionfocused. The tool should display a simple risk flag, the main abnormal findings, and no more than one short explanation line.

Inconsistent or missing vital signs capture	Some triage records may have missing, delayed, or inconsistent vital signs, especially when the department is crowded or when the patient is unstable.	Missing inputs can produce unreliable AI output or false reassurance.	Build missing-data handling into the system. If key inputs are absent, the tool should show "insufficient data for reliable flag" instead of giving an overconfident prediction.

These constraints are important because they determine whether the proposed AI system would bring clinical value or create additional workflow burden. Literature on resourcelimited triage implementation shows that needs assessment, electronic data systems, staff capacity, and local workflow design are central to triage improvement (Mitchell et al., 2024). Reviews of AI triage also identify data quality, staff trust, equipment limitations, and implementation challenges as major barriers (Da'Costa et al., 2025; Tahernejad et al., 2024). In low-resource settings, machine learning may improve triage prediction, but further validation and practical integration are still needed before clinical use (Nsubuga et al., 2025).

8.2 Clinical Stakeholders

Stakeholder	**Role in Workflow**	**Main Concern**	**What Success Looks Like**
Sister Patrice Alleyne, ED Nursein-Charge / Triage Nurse	Leads or oversees triage assessment and patient prioritisation	The tool must not slow down triage, create alert fatigue, or undermine nurse judgement	A short, useful risk flag that supports nurse decisions without replacing them
Dr. De Fretias, ED Consultant / Project Lead	Reviews high-risk patients and supports clinical governance	The tool must be safe, validated, and clinically defensible before any real use	A pilot that clearly reports false alerts, missed high-risk cases, and staff feedback
Registration/Admin Clerk	Records demographic and administrative	The tool must not depend on perfect registration data	A workflow where essential clinical flags can still occur
	information at arrival	before early risk identification	even if registration is incomplete

Porter / Support Staff	Helps move patients between triage, waiting areas, treatment areas, and other departments	The tool should not create unclear routing instructions or extra confusion during patient movement	Clear communication from nursing staff about which patients need faster movement
Patient Advocate	Represents patient safety, fairness, and communication concerns	Patients should not be disadvantaged by missing data, bias, or unclear AI decisions	A transparent system where clinicians remain responsible and patients are not managed by AI alone

9. Risk Analysis

The proposed AI-assisted triage system carries manageable risk only if deployment is gradual, locally validated, and clinically supervised. It should not be used as an autonomous triage decision-maker. The safest design is human-in-the-loop: the AI flags possible risk, but the triage nurse remains responsible for the final triage decision.

9.1 Draft Risk Register

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
Distribution Shift Across Hospitals	AI-Technical	M — performance may change at another hospital or during outbreaks.	H — high-risk patients may be missed.	Silent validation at each new site before go-live; pause rollout if Level 1/2 recall drops by >3 percentage points.	Site recall and calibration stay within ±3 percentage points of baseline.
Data Leakage from Post-Triage Variables	AI-Technical	M — dataset may contain variables only known after triage.	H — model may look accurate in testing but fail in real use.	Audit all features before deployment; remove any input not available at the time of triage.	100% of approved features are timestamped before triage decision.
Alert Fatigue	Operational	H — frequent alerts may train nurses to ignore the system.	H — true urgent alerts may be missed.	Review alert rate weekly; adjust threshold if fewer than 10% of alerts lead to clinical action.	Weekly alertaction rate remains above 10%.

EHR Integration or Downtime Failure	Operational	M — EHR delays or downtime may occur.	H — triage may be delayed or incomplete.	Keep paper fallback workflow; run quarterly downtime drills.	100% of triage staff pass downtime drill; fallback starts within 5 minutes.
Automation Bias	Ethical	M — clinicians may trust the AI too much.	H — staff may follow unsafe AI advice over clinical judgement.	Interface states AI is support only; nurses document agreement or override for highrisk alerts.	Staff override unsafe AI recommendations in ≥90% of simulations.
Accountability and Consent Gap	Ethical	M — patients may not know AI supported triage.	H — unclear responsibility may delay investigation after harm.	Name a clinical owner; create incident review protocol; provide patient-facing disclosure.	100% of AI incidents have named reviewer within 48 hours.
Under-Representation	Equity	H — some groups may be under-	H — model may under-triage elderly, rural,	Check performance by age, gender,	Subgroup recall gap remains ≤5
of Patient Groups		represented in training data.	paediatric, or atypical cases.	arrival method, and site before rollout.	percentage points.
Language, Literacy, and Infrastructure Disparity	Equity	M — some users may struggle with English-only alerts or weak infrastructure.	M — system may work better at resourced sites than rural sites.	Use plainlanguage interface; review local language needs; complete site readiness checklist.	100% of sites pass readiness checks; ≥90% of staff pass comprehension check.

9.2 Top 3 Risks in Plain Language

1. Alert Fatigue

The system may send too many warnings, causing nurses to ignore them even when they are correct. This is dangerous because the tool is meant to highlight patients who need urgent review, not create extra noise. The mitigation is to track how many alerts lead to real clinical action each week and adjust the alert threshold if the action rate becomes too low.

2. Distribution Shift Across Hospitals

A model trained on Mercer data may not work the same way at another public hospital. Different hospitals may have different patient populations, documentation habits, staffing patterns, or outbreak patterns. The mitigation is to test the model silently at each new hospital before it is used in live triage decisions.

3. Under-Representation of Patient Groups

If some patient groups are missing or under-represented in the training data, the model may be less safe for them. This is especially important for elderly patients, paediatric patients, rural patients, and patients with atypical symptoms. The mitigation is to check model performance separately for different patient groups and stop rollout if one group is being missed more often.

9.3 Residual Risk

Even with these mitigations, the system may still miss unusual presentations or perform poorly in hospitals with weak data quality. This cannot be fully resolved at the proposal stage. For that reason, the first deployment should be a silent pilot before any AI output is used for live patient-facing triage decisions.

9.4 AI Harm / Near-Miss Case Study: Epic Sepsis Model

This case is a documented AI safety near-miss rather than a single confirmed patient injury. The system was the Epic Sepsis Model, a proprietary prediction model used to identify hospitalised patients at risk of sepsis. Sepsis is time-sensitive, so a tool that misses high-risk patients or sends too many false alerts can create real clinical danger. In an external validation study at Michigan Medicine, the model performed worse than expected. The study found poor discrimination and calibration, and at the evaluated threshold the model failed to identify many patients who developed sepsis while also producing a large number of alerts.

The immediate failure was that the model did not reliably separate patients who would develop sepsis from those who would not. In plain language, the system was not accurate enough for the clinical task it was supporting. It missed many sepsis cases and produced alerts that could add work for clinicians. For an emergency department triage or early-warning environment, this matters because nurses and doctors already work under time pressure. A weak alerting system can create two safety problems at the same time: missed deterioration and alert fatigue.

The deeper root cause was not simply that the model "made wrong predictions." The deeper design assumption was that a proprietary model developed elsewhere could be widely implemented before strong independent local validation. The model was embedded in a large electronic health record system, but its performance was not adequately proven across every hospital workflow and patient population where it might be used. This is the same risk the Mercer proposal must avoid. An AI triage model trained at one site may not safely generalise to another hospital without local testing.

This failure is both technical and human-factors related. It is technical because the model's prediction performance and calibration were weak in the validation setting. It is also human-factors related because alerts do not operate in isolation: they enter a real clinical workflow where staff may become overloaded, ignore frequent alerts, or assume that a system built into the EHR has already been fully validated.

A safeguard that would have caught this earlier is mandatory external validation before live clinical use. For the proposed AI-assisted ED triage system, the equivalent control would be a silent pilot at every new hospital. During the silent pilot, the model would generate risk flags in the background while clinicians continue normal triage. The Clinical AI Unit would compare the model's Level 1 and Level 2 recall, false-alert rate, and subgroup performance against predefined thresholds. If the model misses too many high-risk patients or produces too many low-value alerts, deployment should pause until the model is recalibrated or redesigned.

The main lesson is that "implemented" does not mean "safe." A model can be widely used and still be unsafe for a specific hospital unless its performance is checked in that exact setting (Wong et al., 2021).

10. Pilot Evaluation Plan

The success of the 12-week pilot would be judged using both technical and workflow measures. The goal is not only to test whether the model can identify high-risk patients, but also whether the output is usable and safe in the triage environment. **Technical Measures**

1. **Flagging performance:** How well the tool identifies high-risk patients using agreed proxy outcomes such as urgent review, admission, resuscitation area placement, or escalation of care.
2. **False-alert burden:** How many patients are flagged who do not require urgent review.
3. **Missing-data handling:** How often the tool cannot generate a reliable output because key inputs are missing.
4. **Comparison with rule-based thresholds:** Whether the AI model adds value beyond simple vital-sign thresholds. **Workflow Measures**

1. **Timing of the flag:** Whether the risk flag appears before or during the triage nurse's decision, rather than after the decision has already been made.
2. **Staff interpretability:** Whether triage staff can understand why the patient was flagged.
3. **Nurse usefulness rating:** Whether triage nurses feel the output is useful, clear, and not disruptive.
4. **Click burden:** Whether the tool adds unnecessary steps to the triage process.
5. **Safety review:** Whether any high-risk patients were missed or whether any alerts would have created unsafe confusion.

A successful pilot would show that the tool can identify potentially high-risk patients while keeping alert burden manageable, handling missing data transparently, and supporting rather than replacing the triage nurse's judgement.

11. Refined Proposed Pilot Timeline Week Activity

Week	Activity
Weeks 1–2	Confirm workflow map, stakeholder concerns, input variables, and safety rules.
Weeks 3–4	Prepare de-identified, simulated, or Mercer-style triage dataset.
Weeks 5–6	Clean data and check missing, unrealistic, or inconsistent values.
Weeks 7–8	Develop rule-based baseline and simple AI-assisted risk-flagging prototype.
Weeks 9–10	Evaluate flagging performance, false alerts, missing-data handling, and workflow timing.
Week 11	Collect staff-facing feedback using a short review form or structured discussion.
Week 12	Produce prototype summary, evaluation report, and recommendation for next steps.

12. Expected Deliverables

The expected deliverables for the 12-week pilot are:

1. A cleaned de-identified or simulated triage dataset
2. A simple AI-assisted risk-flagging prototype
3. A workflow diagram showing where the tool fits into triage
4. A short evaluation report covering performance, false alerts, missing data, and staff usefulness
5. A recommendation on whether the idea should move forward to a larger validation study

13. Conclusion

This proposal focuses on developing a simple AI-assisted risk-flagging tool for emergency department triage in a resource-constrained setting. The literature suggests that AI and machine learning can support emergency triage, but the main challenge is not only technical performance. The tool must be locally validated, understandable to nurses, and integrated into the existing triage workflow without increasing alert burden.

The Week 3 workflow analysis shows that the most realistic plug-in points are after vitalsign capture and during the triage nurse assessment, with a possible secondary deterioration screen after triage is completed. The main constraints are electronic health record data lag, limited triage nurse time, and inconsistent or missing vital signs capture. By designing around these constraints and keeping the nurse as the final decision-maker, the proposed pilot remains realistic, safe, and clinically grounded.

References

Agius, S., Cassar, V., Magri, C., Khan, W., Obe, D. A.-J., Caruana, G., & Topham, L. (2025). Predicting Emergency Severity Index (ESI) level, hospital admission, and admitting ward in an emergency department using data-driven machine learning. *BMC Medical Informatics and Decision Making*, *25*(1), 281. https://doi.org/10.1186/s12911-025-02941-9

Aljubran, H., Aljubran, M., AlAwami, A., Aljubran, M., Alkhalifah, M., Alkhalifah, M., Alkhalifah, A., & Alabdullah, T. (2025). Examining the Use of Machine Learning Algorithms to Enhance the Pediatric Triaging Approach. *Open Access Emergency Medicine*, *Volume 17*, 51–61. https://doi.org/10.2147/OAEM.S494280

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. *Procedia Computer Science*, *251*, 430–437. https://doi.org/10.1016/j.procs.2024.11.130

Ciecierski-Holmes, T., Singh, R., Axt, M., Brenner, S., & Barteit, S. (2022). Artificial intelligence for strengthening healthcare systems in low- and middle-income countries: A systematic scoping review. *Npj Digital Medicine*, *5*(1), 162. https://doi.org/10.1038/s41746-022-00700-y

Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*, *197*, 105838. https://doi.org/10.1016/j.ijmedinf.2025.105838

De Freitas, L., Goodacre, S., O'Hara, R., Thokala, P., & Hariharan, S. (2020). Qualitative exploration of patient flow in a Caribbean emergency department. *BMJ Open*, *10*(12), e041422. https://doi.org/10.1136/bmjopen-2020-041422

Ethics and Governance of Artificial Intelligence for Health: WHO Guidance (1st ed). (2021). World Health Organization.

Fuller, G. W., Hasan, M., Hodkinson, P., McAlpine, D., Goodacre, S., Bath, P. A., Sbaffi, L., Omer, Y., Wallis, L., & Marincowitz, C. (2023). Training and testing of a gradient boosted machine learning model to predict adverse outcome in patients presenting to emergency departments with suspected covid-19 infection in a middle-income setting. *PLOS Digital Health*, *2*(9), e0000309. https://doi.org/10.1371/journal.pdig.0000309

Mallon, O., Lippert, F., & Pilot, E. (2025). Artificial intelligence in prehospital emergency care systems in low- and middle-income countries: Cure or curiosity? Insights from a qualitative study. *Frontiers in Public Health*, *13*, 1632029. https://doi.org/10.3389/fpubh.2025.1632029

Masanneck, L., Schmidt, L., Seifert, A., Kölsche, T., Huntemann, N., Jansen, R., Mehsin, M., Bernhard, M., Meuth, S. G., Böhm, L., & Pawlitzki, M. (2024). Triage Performance Across Large Language Models, ChatGPT, and Untrained Doctors in Emergency Medicine: Comparative Study. *Journal of Medical Internet Research*, *26*, e53297. https://doi.org/10.2196/53297

Mitchell, R., White, L., Elton, L., Luke, C., Bornstein, S., & Atua, V. (2024). Triage implementation in resource-limited emergency departments: Sharing tools and experience from the Pacific region. *International Journal of Emergency Medicine*, *17*(1), 21. https://doi.org/10.1186/s12245-024-00583-8

Nsubuga, M., Kintu, T. M., Please, H., Stewart, K., & Navarro, S. M. (2025). Enhancing trauma triage in low-resource settings using machine learning: A performance comparison with the Kampala Trauma Score. *BMC Emergency Medicine*, *25*(1), 14. https://doi.org/10.1186/s12873-025-01175-2

Tahernejad, A., Sahebi, A., Abadi, A. S. S., & Safari, M. (2024). Application of artificial intelligence in triage in emergencies and disasters: A systematic review. *BMC Public Health*, *24*(1), 3203. https://doi.org/10.1186/s12889-024-20447-3

Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus*. https://doi.org/10.7759/cureus.59906

Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., Pestrue, J., Phillips, M., Konye, J., Penoza, C., Ghous, M., & Singh, K. (2021). External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients. *JAMA Internal Medicine*, *181*(8), 1065. https://doi.org/10.1001/jamainternmed.2021.2626